Aim: To install MS SQL Server and perform data management operations using SQL Server Management Studio (SSMS).

Theory:

Q1. What is MS SQL Server?

Microsoft SQL Server is a **relational database management system (RDBMS)** developed by Microsoft. It is used to store, retrieve, and manage data efficiently using Structured Query Language (SQL) and T-SQL (Transact-SQL — Microsoft's extension). It supports features like transaction processing, business intelligence, analytics, and high availability.

Q2. What is SSMS and what are its features?

SQL Server Management Studio (SSMS) is a free integrated graphical environment for managing, configuring, and administering any SQL Server infrastructure.

- **Object Explorer** – Browse databases, tables, views, stored procedures, etc.
- **Query Editor** – Write, execute, and debug T-SQL queries with syntax highlighting and IntelliSense.
- **Backup & restore** – Perform and schedule database backups.
- **Security management** – Manage logins, users, roles, and permissions.
- **Import/Export Wizard** – Move data between SQL Server and other formats (Excel, CSV, etc.).

Q3. What is the difference between Left Join, Right Join, and Full Join?

Join Type	Description
LEFT JOIN (or LEFT OUTER JOIN)	Returns all rows from the left table, and matching rows from the right table. If no match, right table columns are `NULL`.
RIGHT JOIN (or RIGHT OUTER JOIN)	Returns all rows from the right table, and matching rows from the left table. If no match, left table columns are `NULL`.
FULL JOIN (or FULL OUTER JOIN)	Returns all rows from both tables. Matches where possible; `NULL` in columns from the table lacking a match.

Q4. Write the basic SQL queries for SELECT, INSERT, UPDATE, and DELETE operations with examples.

Assume a table called Students with columns: `StudentID`, `Name`, `Age`, and `Major`.

1) SELECT is used to retrieve data from a database.

```
Example 1: SELECT * FROM Students; - This gets all columns and all rows from the Students table.
Example 2: SELECT Name, Age FROM Students WHERE Major = 'Computer Science'; - This gets only the Name and Age columns for students whose Major is Computer Science.
```

2) INSERT is used to add new rows to a table.

```
Example 1: INSERT INTO Students (StudentID, Name, Age, Major) VALUES (1, 'Alice Johnson',
20, 'Biology'); - This inserts a single student with specified values for selected
columns.
Example 2: INSERT INTO Students VALUES (2, 'Bob Smith', 22, 'Mathematics'); - This inserts
a new row by providing values for all columns in the order they appear in the table.
```

3) UPDATE is used to modify existing data in a table.

```
Example 1: UPDATE Students SET Major = 'Physics' WHERE Name = 'Bob Smith'; - This changes
Bob Smith's major to Physics.
Example 2: UPDATE Students SET Age = 21, Major = 'Computer Science' WHERE StudentID = 1; -
This updates both Age and Major for the student with StudentID 1.
```

4) DELETE is used to remove rows from a table.

```
Example 1: DELETE FROM Students WHERE StudentID = 2; - This deletes only the student with
StudentID 2.
Example 2: DELETE FROM Students WHERE Age > 25; - This deletes all students older than 25.
Example 3: DELETE FROM Students; - This deletes all rows from the Students table (use with
caution).
```

Q5. What are the steps to create a database and a table in SSMS?

There are two methods: using the graphical interface or using T-SQL queries.

Method 1: Using the GUI (no coding):

First, open SSMS and connect to your SQL Server instance.

To create a database, right-click on the Databases folder in Object Explorer, then select "New Database."

Enter a name for your database, such as UniversityDB. You can optionally set file paths, initial size, and autogrowth settings. Click OK to create the database.

To create a table, expand your newly created database in Object Explorer. Right-click on the Tables folder, then select "New" and then "Table."

A design grid will appear where you define your columns.

For each column, enter a Column Name (e.g., StudentID, Name, Age, Major), choose a Data Type (e.g., int, nvarchar(50), int, nvarchar(50)), and specify whether Allow Nulls is checked. To set a primary key, right-click on the StudentID row and choose "Set Primary Key."

After defining all columns, save the table by pressing Ctrl+S and give it a name like Students. The table will appear under the Tables folder.

Method 2: Using T-SQL Query Editor:

Open a new query window in SSMS by clicking "New Query." Then type and execute the following commands:

```
CREATE DATABASE UniversityDB;
GO
USE UniversityDB;
GO
CREATE TABLE Students (
StudentID INT PRIMARY KEY,
Name NVARCHAR(50) NOT NULL,
Age INT,
Major NVARCHAR(50)
);
GO
```

Q6. Perform the following using MS SQL Server and SSMS:

Install SQL Server and SSMS

Create a database named DMBI_LAB

```
Select SQLCMD
--
-- run below commmand from command prompt
-- sqlcmd -S GHOULED-GADGET -No -E
--

select @@Version
go


------------------------------------------------------------------------------------------------------------------------
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------------------------------------------------------------
Microsoft SQL Server 2025 (RTM) - 17.0.1000.7 (X64)
        Oct 21 2025 12:05:57
        Copyright (C) 2025 Microsoft Corporation
        Standard Developer Edition (64-bit) on Windows 10 Home Single Language 10.0 <X64> (Build 19045: )


(1 rows affected)
1> CREATE DATABASE DMBI_LAB;
2> GO
1> USE DMBI_LAB;
2> GO
Changed database context to 'DMBI_LAB'.
```

Create two tables — Student (ID, Name, Age, Gender, Branch) and Teacher (ID, Name, Branch)

```
SQLCMD
Microsoft SQL Server 2025 (RTM) - 17.0.1000.7 (X64)
        Oct 21 2025 12:05:57
        Copyright (C) 2025 Microsoft Corporation
        Standard Developer Edition (64-bit) on Windows 10 Home Single Language 10.0 <X64> (Build 19045: )


(1 rows affected)
1> CREATE DATABASE DMBI_LAB;
2> GO
1> USE DMBI_LAB;
2> GO
Changed database context to 'DMBI_LAB'.
1> CREATE TABLE Student (
2>     ID INT PRIMARY KEY,
3>     Name VARCHAR(50) NOT NULL,
4>     Age INT,
5>     Gender VARCHAR(10),
6>     Branch VARCHAR(50)
7> );
8> GO
1> CREATE TABLE Teacher (
2>     ID INT PRIMARY KEY,
3>     Name VARCHAR(50) NOT NULL,
4>     Branch VARCHAR(50)
5> );
6> GO
```

Insert data into both tables

```
SQLCMD
3>     Name VARCHAR(50) NOT NULL,
4>     Branch VARCHAR(50)
5> );
6> GO
1> INSERT INTO Student (ID, Name, Age, Gender, Branch) VALUES
2> (1, 'Rahul Sharma', 20, 'Male', 'Computer Science'),
3> (2, 'Priya Patel', 21, 'Female', 'Information Technology'),
4> (3, 'Amit Kumar', 19, 'Male', 'Computer Science'),
5> (4, 'Neha Singh', 22, 'Female', 'Electronics'),
6> (5, 'Rajesh Verma', 20, 'Male', 'Mechanical'),
7> (6, 'Sneha Reddy', 21, 'Female', 'Computer Science'),
8> (7, 'Vikram Joshi', 23, 'Male', 'Civil'),
9> (8, 'Anjali Nair', 20, 'Female', 'Information Technology');
10> GO

(8 rows affected)
1> INSERT INTO Teacher (ID, Name, Branch) VALUES
2> (101, 'Dr. Suresh Kumar', 'Computer Science'),
3> (102, 'Prof. Meera Iyer', 'Information Technology'),
4> (103, 'Dr. Rajiv Menon', 'Electronics'),
5> (104, 'Prof. Sunita Rao', 'Mechanical'),
6> (105, 'Dr. Anand Desai', 'Mathematics'),
7> (106, 'Prof. Priyanka Das', 'Physics');
8> GO

(6 rows affected)
1>
```

View data in the tables

```
1> SELECT * FROM Student;
2> GO
ID          Name                                               Age         Gender     Branch

----------- -------------------------------------------------- ----------- ---------- ----------------------------------
---------------
          1 Rahul Sharma                                                20 Male       Computer Science

          2 Priya Patel                                                 21 Female     Information Technology

          3 Amit Kumar                                                  19 Male       Computer Science

          4 Neha Singh                                                  22 Female     Electronics

          5 Rajesh Verma                                                20 Male       Mechanical

          6 Sneha Reddy                                                 21 Female     Computer Science

          7 Vikram Joshi                                                23 Male       Civil

          8 Anjali Nair                                                 20 Female     Information Technology


(8 rows affected)
```

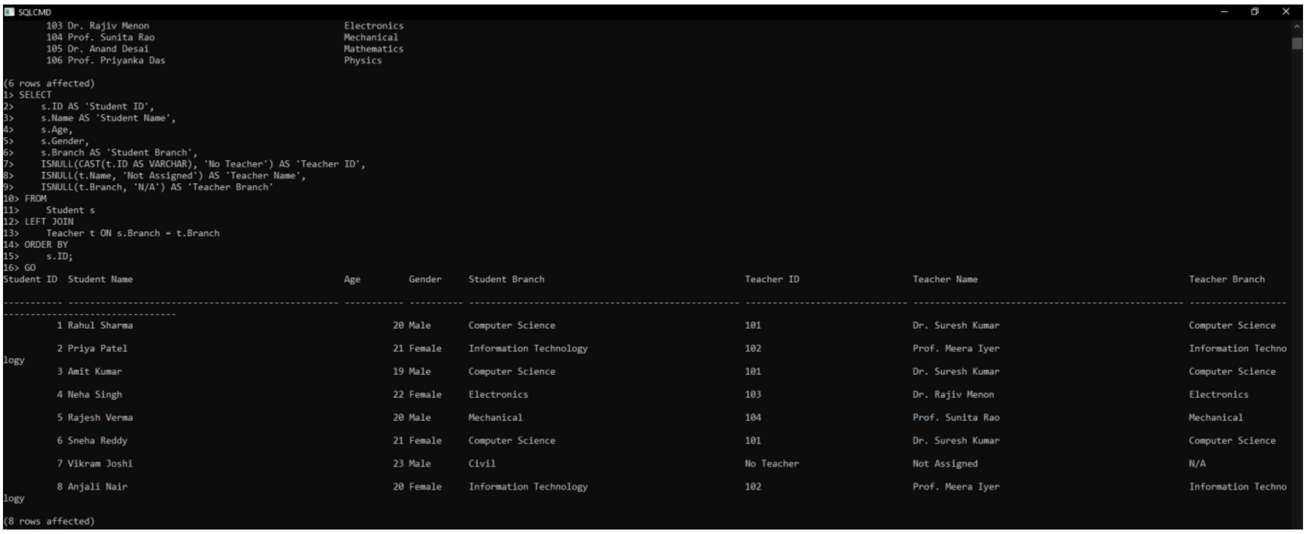
```
1> SELECT * FROM Teacher;
2> GO
ID          Name                                               Branch
----------- -------------------------------------------------- --------------------------------------------------
        101 Dr. Suresh Kumar                                   Computer Science
        102 Prof. Meera Iyer                                   Information Technology
        103 Dr. Rajiv Menon                                    Electronics
        104 Prof. Sunita Rao                                   Mechanical
        105 Dr. Anand Desai                                    Mathematics
        106 Prof. Priyanka Das                                 Physics

(6 rows affected)
```

Perform Left Join, Right Join, and Full Join on the two tables

Left join:

```
SELECT
    s.ID AS 'Student ID',
    s.Name AS 'Student Name',
    s.Age,
    s.Gender,
    s.Branch AS 'Student Branch',
    ISNULL(CAST(t.ID AS VARCHAR), 'No Teacher') AS 'Teacher ID',
    ISNULL(t.Name, 'Not Assigned') AS 'Teacher Name',
    ISNULL(t.Branch, 'N/A') AS 'Teacher Branch'
FROM
    Student s
LEFT JOIN
    Teacher t ON s.Branch = t.Branch
ORDER BY
    s.ID;
GO
```



Student ID	Student Name	Age	Gender	Student Branch	Teacher ID	Teacher Name	Teacher Branch
1	Rahul Sharma	20	Male	Computer Science	101	Dr. Suresh Kumar	Computer Science
2	Priya Patel	21	Female	Information Technology	102	Prof. Meera Iyer	Information Technology
3	Amit Kumar	19	Male	Computer Science	101	Dr. Suresh Kumar	Computer Science
4	Neha Singh	22	Female	Electronics	103	Dr. Rajiv Menon	Electronics
5	Rajesh Verma	20	Male	Mechanical	104	Prof. Sunita Rao	Mechanical
6	Sneha Reddy	21	Female	Computer Science	101	Dr. Suresh Kumar	Computer Science
7	Vikram Joshi	23	Male	Civil	No Teacher	Not Assigned	N/A
8	Anjali Nair	20	Female	Information Technology	102	Prof. Meera Iyer	Information Technology

(8 rows affected)

Right join:

```
SELECT
    ISNULL(CAST(s.ID AS VARCHAR), 'No Student') AS 'Student ID',
    ISNULL(s.Name, 'Not Enrolled') AS 'Student Name',
    ISNULL(s.Branch, 'N/A') AS 'Student Branch',
    t.ID AS 'Teacher ID',
    t.Name AS 'Teacher Name',
    t.Branch AS 'Teacher Branch'
FROM
    Student s
RIGHT JOIN
    Teacher t ON s.Branch = t.Branch
ORDER BY
    t.ID;
GO
```

```
3> SELECT
4>     ISNULL(CAST(s.ID AS VARCHAR), 'No Student') AS 'Student ID',
5>     ISNULL(s.Name, 'Not Enrolled') AS 'Student Name',
6>     ISNULL(s.Branch, 'N/A') AS 'Student Branch',
7>     t.ID AS 'Teacher ID',
8>     t.Name AS 'Teacher Name',
9>     t.Branch AS 'Teacher Branch'
10> FROM
11>     Student s
12> RIGHT JOIN
13>     Teacher t ON s.Branch = t.Branch
14> ORDER BY
15>     t.ID;
16> GO
Shows all teachers and their matching students (if any)
```

Student ID	Student Name	Student Branch	Teacher ID	Teacher Name	Teacher Branch
1	Rahul Sharma	Computer Science	101	Dr. Suresh Kumar	Computer Science
3	Amit Kumar	Computer Science	101	Dr. Suresh Kumar	Computer Science
6	Sneha Reddy	Computer Science	101	Dr. Suresh Kumar	Computer Science
2	Priya Patel	Information Technology	102	Prof. Meera Iyer	Information Technology
8	Anjali Nair	Information Technology	102	Prof. Meera Iyer	Information Technology
4	Neha Singh	Electronics	103	Dr. Rajiv Menon	Electronics
5	Rajesh Verma	Mechanical	104	Prof. Sunita Rao	Mechanical
No Student	Not Enrolled	N/A	105	Dr. Anand Desai	Mathematics
No Student	Not Enrolled	N/A	106	Prof. Priyanka Das	Physics

Full join:

```
SELECT
    s.ID AS 'Student ID',
    s.Name AS 'Student Name',
    s.Branch AS 'Student Branch',
    t.ID AS 'Teacher ID',
    t.Name AS 'Teacher Name',
    t.Branch AS 'Teacher Branch',
    CASE
        WHEN s.ID IS NOT NULL AND t.ID IS NOT NULL THEN 'Matched'
        WHEN s.ID IS NOT NULL AND t.ID IS NULL THEN 'No Teacher Assigned'
        ELSE 'No Students Enrolled'
    END AS 'Status'
FROM
    Student s
FULL JOIN
    Teacher t ON s.Branch = t.Branch
ORDER BY
    ISNULL(s.ID, 999), ISNULL(t.ID, 999);
GO
```

```
1> SELECT
2>     s.ID AS 'Student ID',
3>     s.Name AS 'Student Name',
4>     s.Branch AS 'Student Branch',
5>     t.ID AS 'Teacher ID',
6>     t.Name AS 'Teacher Name',
7>     t.Branch AS 'Teacher Branch',
8>     CASE
9>         WHEN s.ID IS NOT NULL AND t.ID IS NOT NULL THEN 'Matched'
10>         WHEN s.ID IS NOT NULL AND t.ID IS NULL THEN 'No Teacher Assigned'
11>         ELSE 'No Students Enrolled'
12>     END AS 'Status'
13> FROM
14>     Student s
15> FULL JOIN
16>     Teacher t ON s.Branch = t.Branch
17> ORDER BY
18>     ISNULL(s.ID, 999), ISNULL(t.ID, 999);
19> GO
```

Student ID	Student Name	Student Branch	Teacher ID	Teacher Name	Teacher Branch	Status
1	Rahul Sharma	Computer Science	101	Dr. Suresh Kumar	Computer Science	Matched
2	Priya Patel	Information Technology	102	Prof. Meera Iyer	Information Technology	Matched
3	Amit Kumar	Computer Science	101	Dr. Suresh Kumar	Computer Science	Matched
4	Neha Singh	Electronics	103	Dr. Rajiv Menon	Electronics	Matched
5	Rajesh Verma	Mechanical	104	Prof. Sunita Rao	Mechanical	Matched
6	Sneha Reddy	Computer Science	101	Dr. Suresh Kumar	Computer Science	Matched
7	Vikram Joshi	Civil	NULL	NULL	NULL	No Teacher Assigned
8	Anjali Nair	Information Technology	102	Prof. Meera Iyer	Information Technology	Matched
NULL	NULL	NULL	105	Dr. Anand Desai	Mathematics	No Students Enrolled

Conclusion:

Successfully installed MS SQL Server and perform data management operations using SQL Server Management Studio (SSMS).